

GuardianGarment: Enhanced Safety System Specification

Concept Overview: The GuardianGarment is a high-reliability, wearable safety system for individuals prone to wandering. This enhanced specification incorporates a dynamic, high-visibility "Pulsing Alert" mechanism and a descriptive emergency message to ensure clear communication with bystanders.

1. System Architecture

The system relies on an autonomous electronic module (The "Smart Patch") that integrates "Guard" (detection) and "Find" (tracking) capabilities.

The Logic Controller

The ESP32-C3 microcontroller operates in a dual-concurrent state:

- **Guard Mode:** Continuously monitors for the caregiver's BLE whitelist signal.
- **Find Mode (Beacon):** Automatically activates high-strength broadcasting for triangulation if separation occurs.

2. Enhanced Visual Alert: Pulsing Stencil

The system utilizes a back-lit stencil to provide a clear, prioritized alert without digital screen complexity.

- **Pulsing Logic:** Upon trigger, the LED array initiates a "breathing" or "pulsing" pattern. This visual effect is cognitively prioritized by observers as an emergency warning, ensuring the device stands out in busy environments.
- **Message Layout:** The message is formatted into a high-visibility, two-row stencil:
Line 1: "PERSON WITH MEMORY LOSS – PLEASE HELP"
Line 2: "CALL: [MOBILE NUMBER]"
- **Illumination:** A high-intensity LED array positioned behind a diffusion layer ensures the text is legible even in varying light conditions.

3. Active Recovery: BLE "Find" Functionality

The system bridges the gap between passive identification and active recovery:

- **Proximity Guard:** A "Debounce" timer (60–120 seconds) filters out transient signal drops.
- **Beacon Triangulation:** When triggered, the device broadcasts a high-strength signal. Caregivers can use any standard BLE scanner application as a "hot-cold" locator to triangulate the patient's position.

4. Design & Ergonomics

The electronics are housed in a modular, low-profile pouch system designed for comfort and durability.

- **Form Factor:** All components are contained in a slim, waterproof TPU pouch.
- **Modularity:** The module attaches to clothing via a hook-and-loop (Velcro) base, ensuring it can be easily removed for garment maintenance or battery recharging.
- **Thermal Safety:** The system utilizes cool-running LEDs, ensuring the patch remains safe and comfortable against the wearer's clothing.

5. Operational Logic

1. **Stable State:** Caregiver is within range. BLE Handshake active. LEDs off.
2. **Trigger Event:** Connection lost for >60 seconds.
3. **Alert State:** LEDs pulse, revealing the stencil with the alert message. Beacon mode active for tracking.
4. **Recovery:** Caregiver locates the patient via RSSI signal strength monitoring.